

Course Project

REE 305 – Wind Energy Technology

Analysis of December 2022 Wind-Speed Measurements at the Hail Site

Objective

- Implement the wind-energy formulas developed in the course lectures on a real measured data set.
- Sort, bin, and statistically analyse wind-speed data using a spreadsheet, and reproduce the results analytically.
- Compare the discrete (bin/histogram) approach with the continuous Weibull and Rayleigh probability models.

Data Set

The data used in this project is the wind speed recorded at the Hail measurement site during the month of **December 2022**. The anemometer, mounted at a reference height of 10 m, logged one sample every ten minutes, giving

$$N = 31 \text{ days} \times 24 \frac{\text{h}}{\text{day}} \times 6 \frac{\text{samples}}{\text{h}} = 4464 \text{ samples}, \quad T = 31 \times 24 = 744 \text{ hours}.$$

The recorded speeds range from 0 m/s to 8.75 m/s; 323 of the readings (7.24 %) are exact calms ($U = 0$). All calculations below are performed on these 4464 readings.

Question 1

1) Mean Wind Speed (Simple Method)

The mean speed can be obtained by two approaches: the *simple* average and the *bin* (frequency) method. The simple approach adds every reading and divides by their number:

$$\bar{U} = \frac{1}{N} \sum_{i=1}^N U_i = \frac{1}{4464} (0 + 0 + \dots + 2.90) = \boxed{2.126 \text{ m/s}}.$$

2) Standard Deviation

The standard deviation σ is required later for the Weibull shape factor. Using the sample estimator (STDEV.S in Excel):

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (U_i - \bar{U})^2} = \sqrt{\frac{1}{4464-1} \sum_{i=1}^{4464} (U_i - 2.126)^2} = \boxed{1.347 \text{ m/s}}.$$

3) Velocity Histogram

The velocity data are grouped into bins of width 0.5 m/s. Because a large number of readings are exact calms, a dedicated “calm” bin ($U = 0$) is kept separate from the first moving-air bin, exactly as done in the lecture. The number of occurrences in each bin is obtained with the `COUNTIF` function. The mean speed recomputed from the bins,

$$\bar{U}_{\text{bin}} = \sum_i \left(\frac{f_i}{N} \right) \bar{U}_i = 2.104 \text{ m/s},$$

agrees with the simple mean of Part 1 to within 1 %, confirming the binning.

Table 1: Velocity histogram (bin width 0.5 m/s) for December 2022.

Bin	Min U	Max U	Avg U	Freq.	%Freq	%Freq· U
1	0.00	0.00	0.00	323	0.0724	0.0000
2	0.00	0.50	0.25	117	0.0262	0.0066
3	0.50	1.00	0.75	393	0.0880	0.0660
4	1.00	1.50	1.25	696	0.1559	0.1949
5	1.50	2.00	1.75	898	0.2012	0.3520
6	2.00	2.50	2.25	574	0.1286	0.2893
7	2.50	3.00	2.75	459	0.1028	0.2828
8	3.00	3.50	3.25	386	0.0865	0.2810
9	3.50	4.00	3.75	216	0.0484	0.1815
10	4.00	4.50	4.25	140	0.0314	0.1333
11	4.50	5.00	4.75	120	0.0269	0.1277
12	5.00	5.50	5.25	51	0.0114	0.0600
13	5.50	6.00	5.75	39	0.0087	0.0502
14	6.00	6.50	6.25	30	0.0067	0.0420
15	6.50	7.00	6.75	6	0.0013	0.0091
16	7.00	7.50	7.25	7	0.0016	0.0114
17	7.50	8.00	7.75	3	0.0007	0.0052
18	8.00	8.50	8.25	4	0.0009	0.0074
19	8.50	∞	8.75	2	0.0004	0.0039
Total	–	–	–	4464	1.0000	2.1042

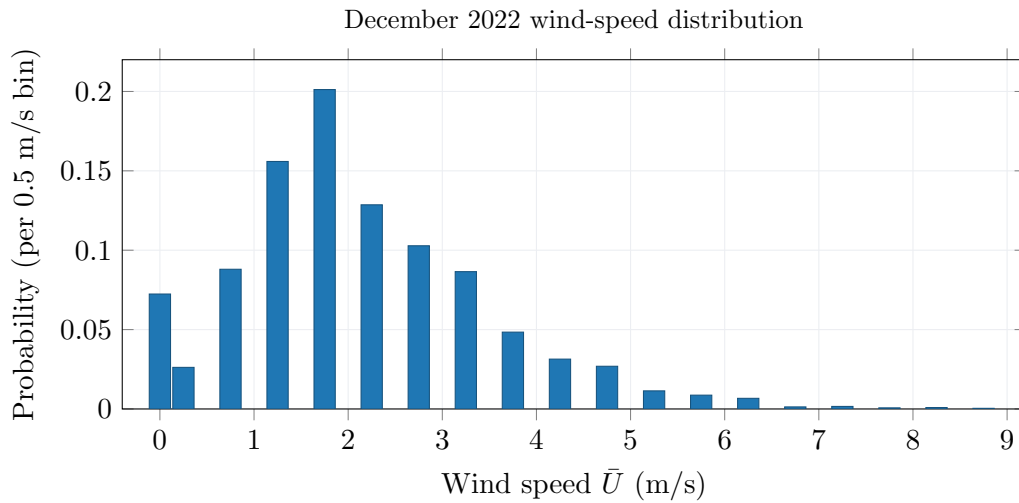


Figure 1: Relative frequency histogram of the measured wind speed.

4) Velocity–Duration Curve

Using the histogram, the frequency is accumulated in reverse to obtain the fraction of the month during which the wind speed *equals or exceeds* a given value. Multiplying this reversed cumulative frequency by the number of hours in the month ($T = 744$ h) gives the number of hours per month:

$$\text{Hours}(\bar{U}_i) = \left(\sum_{j \geq i} \frac{f_j}{N} \right) \times 744.$$

Table 2: Velocity–duration data (bin method).

Bin	Avg U	%Freq	Cum. Freq (rev.)	Hours/month
1	0.00	0.0724	1.0000	744.0
2	0.25	0.0262	0.9276	690.2
3	0.75	0.0880	0.9014	670.7
4	1.25	0.1559	0.8134	605.2
5	1.75	0.2012	0.6575	489.2
6	2.25	0.1286	0.4563	339.5
7	2.75	0.1028	0.3277	243.8
8	3.25	0.0865	0.2249	167.3
9	3.75	0.0484	0.1384	103.0
10	4.25	0.0314	0.0901	67.0
11	4.75	0.0269	0.0587	43.7
12	5.25	0.0114	0.0318	23.7
13	5.75	0.0087	0.0204	15.2
14	6.25	0.0067	0.0116	8.7
15	6.75	0.0013	0.0049	3.7
16	7.25	0.0016	0.0036	2.7
17	7.75	0.0007	0.0020	1.5
18	8.25	0.0009	0.0013	1.0
19	8.75	0.0004	0.0004	0.3

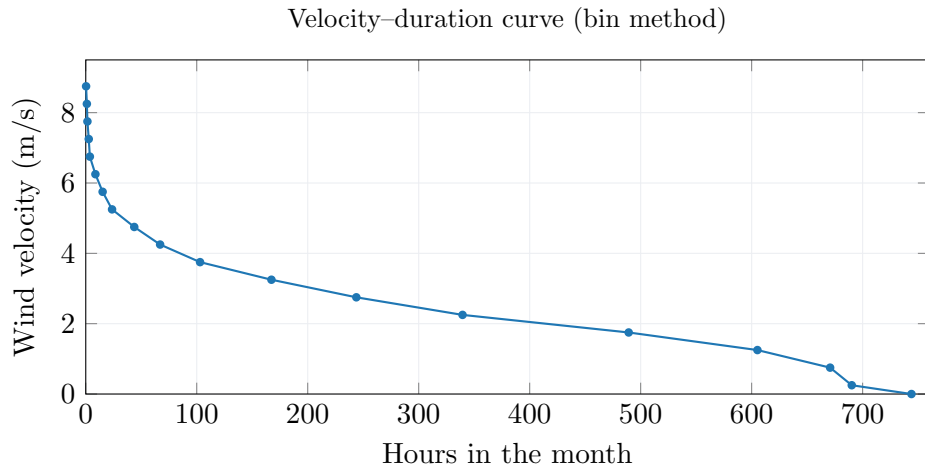


Figure 2: Number of hours per month for which the wind speed is exceeded.

5) Available Energy at 10 m

The available wind energy per unit swept area during the month is the sum, over all bins, of the power density times the number of hours the wind blows in that bin:

$$\text{Power density} = \frac{1}{2} \rho \bar{U}_i^3 \quad (\rho = 1.225 \text{ kg/m}^3), \quad E = \sum_i \left(\frac{1}{2} \rho \bar{U}_i^3 \right) \times \text{Hours}_i.$$

The total available energy at the 10 m measurement height is

$$E_{10} = 10441.7 \text{ Wh/m}^2 = 10.442 \text{ kWh/m}^2.$$

6) Available Energy at 50 m

The speeds are measured at 10 m; they are extrapolated to a hub height of 50 m with the power-law profile using an exponent $\alpha = \frac{1}{7}$:

$$\frac{U_{50}}{U_{10}} = \left(\frac{z_{50}}{z_{10}} \right)^\alpha = \left(\frac{50}{10} \right)^{1/7} = 1.2585.$$

Because the power density scales with the *cube* of the speed,

$$\frac{P_{50}}{P_{10}} = \left(\frac{U_{50}}{U_{10}} \right)^3 = \left(\frac{z_{50}}{z_{10}} \right)^{3\alpha} = 5^{3/7} = 1.9932.$$

Hence the available energy at 50 m is $E_{50} = E_{10} \times 1.9932$:

$$E_{50} = 20812.8 \text{ Wh/m}^2 = 20.813 \text{ kWh/m}^2.$$

Raising the tower from 10 m to 50 m therefore roughly *doubles* the harvestable energy.

Table 3: Available energy at 10 m and 50 m (per 0.5 m/s bin).

Bin	Avg U (m/s)	Hours	$\bar{U}^3$	PD_{10} (W/m ²)	E_{10} (Wh/m ²)	PD_{50} (W/m ²)	E_{50} (Wh/m ²)
1	0.00	53.8	0.00	0.00	0.0	0.00	0.0
2	0.25	19.5	0.02	0.01	0.2	0.02	0.4
3	0.75	65.5	0.42	0.26	16.9	0.52	33.7
4	1.25	116.0	1.95	1.20	138.8	2.38	276.6
5	1.75	149.7	5.36	3.28	491.3	6.54	979.3
6	2.25	95.7	11.39	6.98	667.4	13.91	1330.4
7	2.75	76.5	20.80	12.74	974.5	25.39	1942.3
8	3.25	64.3	34.33	21.03	1352.7	41.91	2696.2
9	3.75	36.0	52.73	32.30	1162.8	64.38	2317.7
10	4.25	23.3	76.77	47.02	1097.1	93.72	2186.8
11	4.75	20.0	107.17	65.64	1312.9	130.84	2616.8
12	5.25	8.5	144.70	88.63	753.4	176.66	1501.6
13	5.75	6.5	190.11	116.44	756.9	232.10	1508.6
14	6.25	5.0	244.14	149.54	747.7	298.06	1490.3
15	6.75	1.0	307.55	188.37	188.4	375.47	375.5
16	7.25	1.2	381.08	233.41	272.3	465.24	542.8
17	7.75	0.5	465.48	285.11	142.6	568.29	284.1
18	8.25	0.7	561.52	343.93	229.3	685.53	457.0
19	8.75	0.3	669.92	410.33	136.8	817.88	272.6
Total					10441.7		20812.8

Question 2

We now model the same data with two continuous probability distributions. The cumulative form is the integral of the probability density, $F(U) = \int_0^U f(U) dU$.

Weibull. The Weibull density and its cumulative distribution are

$$f(U) = \frac{k}{c} \left(\frac{U}{c} \right)^{k-1} e^{-(U/c)^k}, \quad F(U) = 1 - e^{-(U/c)^k},$$

where k is the shape factor and c the scale factor.

Rayleigh. The Rayleigh distribution is the special case of Weibull with $k = 2$; it depends on the mean speed only:

$$f(U) = \frac{\pi}{2} \frac{U}{\bar{U}^2} e^{-\frac{\pi}{4}(U/\bar{U})^2}, \quad c = \frac{2\bar{U}}{\sqrt{\pi}}.$$

a) Distribution Curves

The Weibull parameters are found from the empirical Justus (1978) relations using the mean and standard deviation of Parts 1–2:

$$k = \left(\frac{\sigma}{\bar{U}} \right)^{-1.086} = \left(\frac{1.347}{2.126} \right)^{-1.086} = 1.641, \quad c = \frac{\bar{U}}{\Gamma(1 + \frac{1}{k})} = \frac{2.126}{\Gamma(1.609)} = 2.376.$$

For Rayleigh, $c = \frac{2\bar{U}}{\sqrt{\pi}} = 2.399$.

Note on normalisation. The distributions give a probability *density* $f(U)$. The probability of a 0.5 m/s bin is $f(\bar{U}_i) \Delta U$ with $\Delta U = 0.5$ m/s. Multiplying by the bin width (rather than an arbitrary factor) makes the probabilities sum to unity, as shown by the totals in Table 4.

Table 4: Weibull and Rayleigh distributions evaluated at the bin midpoints.

Bin	Avg U	$f_W(U)$	$p_W = f_W \Delta U$	$f_R(U)$	$p_R = f_R \Delta U$
1	0.00	0.0000	0.0000	0.0000	0.0000
2	0.25	0.1589	0.0795	0.0860	0.0430
3	0.75	0.2835	0.1418	0.2364	0.1182
4	1.25	0.3229	0.1614	0.3311	0.1656
5	1.75	0.3099	0.1550	0.3572	0.1786
6	2.25	0.2673	0.1337	0.3244	0.1622
7	2.75	0.2128	0.1064	0.2568	0.1284
8	3.25	0.1587	0.0793	0.1802	0.0901
9	3.75	0.1117	0.0559	0.1132	0.0566
10	4.25	0.0747	0.0374	0.0640	0.0320
11	4.75	0.0477	0.0239	0.0327	0.0164
12	5.25	0.0292	0.0146	0.0152	0.0076
13	5.75	0.0171	0.0086	0.0064	0.0032
14	6.25	0.0097	0.0048	0.0024	0.0012
15	6.75	0.0053	0.0026	0.0009	0.0004
16	7.25	0.0028	0.0014	0.0003	0.0001
17	7.75	0.0014	0.0007	0.0001	0.0000
18	8.25	0.0007	0.0003	0.0000	0.0000
19	8.75	0.0003	0.0002	0.0000	0.0000
Sum			1.0073		1.0036

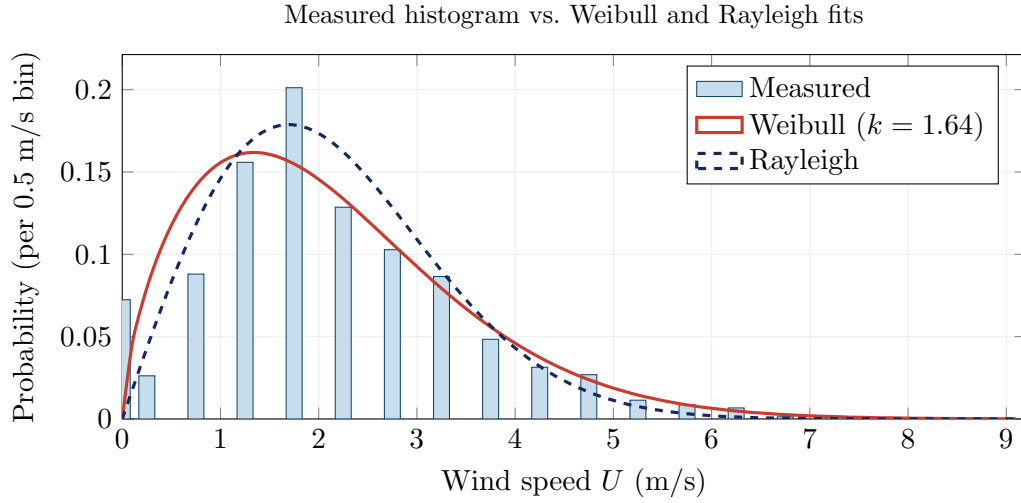


Figure 3: Weibull and Rayleigh probability curves superimposed on the measured histogram.

b) Weibull and Rayleigh Velocity–Duration Curves

Following the same procedure as Part 1.4, the number of hours the wind speed is exceeded is obtained from the reversed cumulative distribution:

$$\text{Hours}(U) = [1 - F(U)] \times T = e^{-(U/c)^k} \times 744.$$

Table 5: Weibull and Rayleigh velocity–duration data.

Bin	Avg U	Hours (Weibull)	Hours (Rayleigh)
1	0.00	744.0	744.0
2	0.25	725.8	736.0
3	0.75	640.0	674.7
4	1.25	525.1	567.1
5	1.75	406.2	437.0
6	2.25	298.2	308.7
7	2.75	208.8	199.9
8	3.25	139.8	118.7
9	3.75	89.8	64.6
10	4.25	55.4	32.2
11	4.75	33.0	14.7
12	5.25	18.9	6.2
13	5.75	10.5	2.4
14	6.25	5.6	0.8
15	6.75	2.9	0.3
16	7.25	1.5	0.1
17	7.75	0.7	0.0
18	8.25	0.3	0.0
19	8.75	0.2	0.0

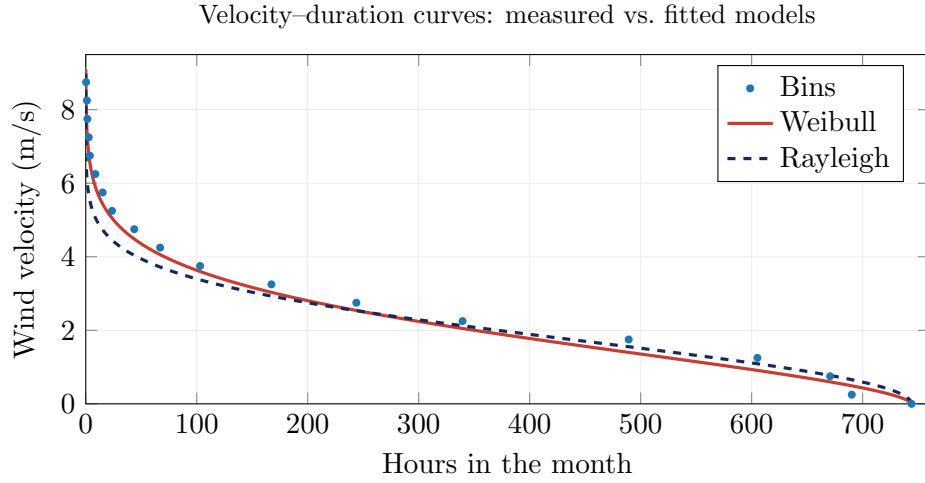


Figure 4: Weibull and Rayleigh duration curves compared with the bin-based curve of Part 1.4.

c) Available Energy from the Weibull Model

Using the Weibull probability of each bin to obtain the hours, the available energy is computed exactly as in Part 1.5:

$$E_W = \sum_i \left(\frac{1}{2} \rho \bar{U}_i^3 \right) \times (p_{W,i} T) = \boxed{10440.9 \text{ Wh/m}^2 = 10.441 \text{ kWh/m}^2}.$$

Table 6: Available energy at 10 m from the Weibull distribution.

Bin	Avg U	p_W	Hours	$PD \text{ (W/m}^2\text{)}$	$E \text{ (Wh/m}^2\text{)}$
1	0.00	0.0000	0.0	0.00	0.0
2	0.25	0.0795	59.1	0.01	0.6
3	0.75	0.1418	105.5	0.26	27.3
4	1.25	0.1614	120.1	1.20	143.7
5	1.75	0.1550	115.3	3.28	378.4
6	2.25	0.1337	99.4	6.98	693.8
7	2.75	0.1064	79.2	12.74	1008.6
8	3.25	0.0793	59.0	21.03	1241.0
9	3.75	0.0559	41.6	32.30	1342.3
10	4.25	0.0374	27.8	47.02	1307.2
11	4.75	0.0239	17.7	65.64	1165.1
12	5.25	0.0146	10.8	88.63	961.6
13	5.75	0.0086	6.4	116.44	741.2
14	6.25	0.0048	3.6	149.54	537.2
15	6.75	0.0026	2.0	188.37	368.0
16	7.25	0.0014	1.0	233.41	239.3
17	7.75	0.0007	0.5	285.11	148.2
18	8.25	0.0003	0.3	343.93	87.7
19	8.75	0.0002	0.1	410.33	49.7
Total					10440.9

Notes on the Analysis

- The month of December has **31 days**, so the correct number of hours is $T = 744$ h (4464 ten-minute samples), and this value is used throughout.
- The Weibull and Rayleigh densities are converted to bin probabilities by multiplying by the bin width $\Delta U = 0.5$ m/s; the columns then sum to ≈ 1 with no ad-hoc scaling.
- In the height extrapolation the power density is scaled by the *cube* of the velocity ratio, $(U_{50}/U_{10})^3 = 5^{3/7} = 1.9932$, i.e. the energy roughly doubles between 10 m and 50 m.

Conclusion

This project applied the wind-energy formulas from the course to a full month of measured data at the Hail site. The mean wind speed for December 2022 is $\bar{U} = 2.126$ m/s with a standard deviation of $\sigma = 1.347$ m/s. The simple average and the bin-method average agree to within 1 %, and the histogram is well represented by a Weibull distribution with $k = 1.641$ and $c = 2.376$, and by a Rayleigh distribution with $c = 2.399$.

The available energy obtained from the discrete bin method, $E_{10} = 10.442$ kWh/m², is essentially identical to that predicted by the continuous Weibull model, $E_W = 10.441$ kWh/m² (a difference of only -0.008%), which validates the statistical approach that needs little more than the mean and standard deviation. Extrapolating to a 50 m hub height nearly doubles the resource to $E_{50} = 20.813$ kWh/m². Both the bin and the fitted models slightly over-predict the mid-range speeds (2–5 m/s) relative to the raw histogram, which is the expected behaviour of these smooth two-parameter distributions.